Phuoc Quyen (Art) Nguyen

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Education

- Clarkson University - Potsdam, New York
- Bachelor of Science in Computer Engineering | Graduation: Dec 2025 | GPA: 3.12
- Master of Science in Electrical & Computer Engineering | Expected Graduation: Dec 2026

Skills

- **Languages**: Python, Java, Kotlin, C, C++, C#, JavaScript (TS/Node.js), SQL, HTML/CSS/Tailwind, VHDL, XAML, TCL
- **Frameworks:** Next.js, React, Flask, WPF, Avalonia, ASP.NET, Django, SPFx, ROS2(Humble), Chirpstack, Autoware
- **Key Skills:** API Integration, RF & Serial Communication, DSA, Unit Testing, App Deployment, Documentation
- **Soft Skills:** Problem Solving, Team Focused Development, Competent Communication

Tools

- **IDEs & Dev Tools:** Visual Studio (Code & 2022), Vivado 2019+, STMCube(IDE & ProG), Eclipse IDE, Vitis, Quartus II, Git (GitHub, Bitbucket), SVN, Jira (Agile), Azure DevOps
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis
- **Cloud & Environment:** AWS,(S3,Lambda), Azure, Google Cloud, Firebase, Docker, Apache, VMware
- **Other Tools:** FPGAs(and SOMs), Astral Stack(UV, Ruff, ty), NFIQ2, Remote Desktop, PuTTY, Powershell, AutoDesk Maya, Windows, Linux

Professional Experience

Teaching Assistant, *Clarkson University* — Jan 2026 - Present

- Designed and graded embedded systems labs supporting hands-on learning for 57 undergraduate students.

Assistant Researcher, *Clarkson University AVHBAC Lab* — April 2024 - Jan 2026

- Designed custom **LoRa + STM32** PCBs integrated with **ChirpStack** or smart farming data collection.
- Developed FPGA designs on **AMD(Xilinx) Kria** platforms for serial communication and motor control on a robotic hand.
- Integrated a **DPU** to hardware-accelerate **YOLOv7** models, improving real-time object detection on embedded hardware.
- Designed a high-resolution (4389 PPI) infant fingerprint sensor and processing pipeline for biometric research.
- Built a desktop application with a custom OpenCV pipeline for near-infrared iris image capture.
- Developed a WebSocket app for EEG data recording using the Emotiv EPOC X headset.

Software Engineering Intern, *C Speed, LLC* — January 2025 - May 2025

- Designed, developed, and documented an internal HR and accounting application integrating two third-party services to streamline payroll workflows and reduce manual processing by ~**96 hours per month**.
- Rewrote an in-house legacy desktop application in **C#** using **Avalonia**, enabling cross-platform support and improving maintainability.
- Partnered with the Director of Operations to develop a company intranet on **Microsoft SharePoint**, developing custom **SPFx** web parts and automating workflows with Power Automate.

Assistant Researcher, *Clarkson University CAMEL Lab* — June 2025 - October 2025

- Built custom vehicle models and point-cloud/vector maps for autonomous driving simulation in Autoware.
- Integrated Autoware with the Scout 2.0 platform for fully autonomous real-world operation.

IT Assistant and Database Manager, Pediatric Medicine Clinic – Phạm Thị Minh Hồng — June 2019 - August 2021

- Implemented and managed a database system to track daily medicine imports, exports, and medical expense statements for a daily patient load of ~100.

Publications

- **Phuoc Quyen Nguyen**, Rumana Sumi, Jonathan Trusty King, Stephanie Shuckers, Masudul H. Imtiaz, *Design of a High-Resolution Infant Fingerprint Scanner Using a Low-Cost COTS Camera*, Ubiquitous Computing, Electronics & Mobile Communication Conference(UEMCON) 2026 - (Accepted)

Projects

Hardware Accelerated AI Fingerprint Sensor and Authentication system purely on an FPGA

- Built a fingerprint sensor that is able to capture and process fingerprints then hardware accelerates a Siamese neural network model through a deep learning processing unit to authenticate biometric characteristics of a fingerprint to verify user identity entirely hosted on an FPGA.

Personal Website

- Developed a personal portfolio site with React(Typescript) and Next.js, deployed on Vercel.